

A/ TERMS AND CONDITIONS OF USE

GENERAL

THESE DRAWINGS HAVE BEEN PRODUCED FOR THE SOLE USE OF THE PROJECT CLIENT, THEIR APPOINTED AGENTS, ARCHITECTS, CONSULTANTS AND CONTRACTORS. BY THE USE OF THESE DRAWINGS AND SPECIFICATION BY THE APPOINTED AGENTS AND INDIVIDUALS, THEY ACKNOWLEDGE AND AGREE TO THE TERMS AND CONDITIONS OF USE AS NOTED BELOW.

THESE STRUCTURAL DRAWINGS AND SPECIFICATIONS HAVE BEEN PREPARED TO ILLUSTRATE THE STRUCTURAL DESIGN PHILOSOPHY AND INTENT. AS SUCH, THESE DRAWINGS PERTAIN SOLELY TO THE "BUILDING STRUCTURE" AND DO NOT ATTEMPT TO DEPICT, DETAIL, OR DESIGN ELEMENTS OF THE FINISHED BUILDING DESIGN WHICH ARE THE RESPONSIBILITY OF OTHER DESIGN DISCIPLINES.

STRUCTURAL BUILDING FRAMEWORK

THESE STRUCTURAL DRAWINGS DESCRIBE THE STRUCTURAL BUILDING FRAMEWORK ONLY. THE STRUCTURAL FRAMEWORK IS DEFINED AS THE PRIMARY GRAVITY AND LATERAL LOAD RESISTING ELEMENTS OF THE BUILDING AND INCLUDES THE FOLLOWING ITEMS WHICH MAY OR MAY NOT PERTAIN TO THIS SPECIFIC PROJECT:

- REINFORCED CONCRETE ELEMENTS WHICH COMPRISE THE PRIMARY LOAD CARRYING FRAMEWORK FOR THE BUILDING
- COMPOSITE OR NON-COMPOSITE ROOF OR FLOOR/METAL DECKING
- STRUCTURAL STEEL ELEMENTS AS DEFINED BY THE CBC CODE OF STANDARD PRACTICE
- STEEL JOISTS - PRE-CAST CONCRETE SLABS - BEAMS COLUMNS OR WALL ELEMENTS

PROVISION HAS BEEN MADE WITHIN THE DESIGN OF THE STRUCTURAL BUILDING FRAMEWORK TO SUPPORT THE LOADS ARISING FROM NON-STRUCTURAL ELEMENTS. SUCH NON-STRUCTURAL ELEMENTS MAY RELY UPON THE STRUCTURAL BUILDING FRAMEWORK FOR THEIR ULTIMATE SUPPORT. HOWEVER, THE DESIGN, SUPPLY AND INSTALLATION OF AND SUPPORT MECHANISMS ARE THE RESPONSIBILITY OF THE SPECIFIC ARCHITECTURAL OR SERVICE TRADE AND/OR OTHER ENGINEERING DISCIPLINE.

SPECIFIC ARCHITECTURAL AND SERVICE ELEMENTS MAY RELY UPON THE STRUCTURAL BUILDING FRAMEWORK FOR ULTIMATE SUPPORT YET WHOSE DESIGN AND CONTRACTUAL RESPONSIBILITY DOES NOT LIE WITH THE STRUCTURAL DESIGNER. THE DESIGNER AND/OR PROVIDERS OF THE FOLLOWING ELEMENTS, ARE WHOLLY RESPONSIBLE FOR DELIVERING THEIR LOADS TO THE STRUCTURE AS DEFINED BY THESE DRAWINGS:

- EXTERIOR BUILDING CLADDING SYSTEM, INCLUDING WIND RESISTING LIGHTWEIGHT METAL STUD SYSTEMS AND CONCRETE MASONRY WALLS
- NON-LOAD BEARING PARTITION
- ANY AND ALL SLABS, TOPPING AND FINISHES (WHETHER OR NOT THEY CONTAIN REINFORCING MATERIALS) NOT ATTACHED TO THE STRUCTURAL BUILDING FRAMEWORK NOT DEFINED ON THE STRUCTURAL DRAWINGS
- BUILDING SERVICES AND EQUIPMENT INCLUDING ANY AND ALL ELEMENTS AND THEIR SUPPORTING ACCESSORIES
- ELEVATING AND CONVEYING SYSTEMS
- ALL EMBEDDED OR SURFACE APPLIED CONNECTION MEANS NOT INTENDED TO SUPPORT A PORTION OF THE STRUCTURAL BUILDING FRAMEWORK

WHERE APPROPRIATE, THE DESIGN AND INSTALLATION OF SUCH ELEMENTS SHALL BE PROVIDED BY A PROFESSIONAL ENGINEER, LICENSED WITHIN THE RELEVANT JURISDICTION. AS THE STRUCTURAL DESIGNER, QUINN DRESSSEL ASSOCIATES IS NOT RESPONSIBLE FOR THE DESIGN OF SUCH ELEMENTS, NOR THE METHOD OR MEANS OF CONNECTING SUCH ELEMENTS TO THE BUILDING STRUCTURE.

NOTWITHSTANDING THE ABOVE, QUINN DRESSSEL ASSOCIATES SHALL HAVE SOLE DISCRETION TO SELECT ANY CONNECTION METHOD OR EMBEDDED ITEM WHICH HE JUDGE TO BE DETRIMENTAL TO THE PERFORMANCE OF THE STRUCTURAL BUILDING FRAMEWORK.

SHOP DRAWINGS

IT IS UNDERSTOOD AND ACKNOWLEDGED BY ALL THAT THE CO-ORDINATION AND PREPARATION OF THE SHOP DRAWINGS IS THE RESPONSIBILITY OF THE GENERAL CONTRACTOR, CONSTRUCTION MANAGER OR PROJECT MANAGER.

QUINN DRESSSEL ASSOCIATES WILL RECEIVE AND REVIEW SHOP DRAWINGS PREPARED BY THE STRUCTURAL TRADES. SUCH REVIEW IS FOR THE SOLE PURPOSE OF ASCERTAINING CONFORMANCE WITH THE GENERAL DESIGN CONCEPT, CRITERIA AND INTENT. TYPICALLY THIS WILL INCLUDE THE FOLLOWING:

- TEMPORARY EXCAVATION SHORING PLANS AND DETAILS
- CONCRETE FALSEWORK DRAWINGS
- REINFORCING STEEL SHOP DRAWINGS
- POST-TENSIONING PLANS AND DETAILS
- STRUCTURAL STEEL SHOP DRAWINGS
- METAL DECK SHOP DRAWINGS
- STEEL JOIST SHOP DRAWINGS
- STRUCTURAL PRE-CAST SHOP DRAWINGS

WHERE SO DIRECTED BY THE CONTRACT DOCUMENTS, THE ABOVE SUBMISSION MAY REQUIRE THE SEAL AND SIGNATURE OF A PROFESSIONAL ENGINEER, SO RELEVANT IN THE GOVERNING JURISDICTION. IN SUCH INSTANCES, THE SOLE RESPONSIBILITY FOR THE DESIGN OF SUCH ELEMENTS LIES WITH THAT DESIGNER AND IS OUTSIDE OF THE SCOPE OF SERVICES PROVIDED BY QUINN DRESSSEL ASSOCIATES. QUINN DRESSSEL ASSOCIATES WILL ALSO RECEIVE AND PROVIDE A LIMITED REVIEW OF SHOP DRAWINGS PREPARED BY TRADES OTHER THAN STRUCTURAL. SUCH REVIEW IS UNDERTAKEN SOLELY TO ASCERTAIN THE EFFECTS OF THE WORK OF OTHER TRADES UPON THE STRUCTURAL BUILDING FRAMEWORK, AND IS LIMITED TO A REVIEW OF THE LOADS AND MANNER IN WHICH THE LOADS ARE DELIVERED TO THE STRUCTURAL BUILDING FRAMEWORK. TYPICALLY, DRAWINGS PREPARED BY TRADES OTHER THAN STRUCTURAL WOULD INCLUDE:

- EXTERIOR BUILDING CLADDING SYSTEM, INDICATING LOADS AND ATTACHMENT METHODOLOGY
- BUILDING DRAWINGS PREPARED BY THE MECHANICAL OR ELECTRICAL TRADES
- CORE REQUESTS BY THE MECHANICAL OR ELECTRICAL TRADES
- EQUIPMENT AND/OR PIPING LOADS AND ATTACHMENT METHODOLOGY
- WINDOW WASHING EQUIPMENT AND ANCHOR ATTACHMENTS

DRAWINGS RECEIVED WHICH DO NOT CONTAIN ADEQUATE LOADING INFORMATION WILL BE RETURNED AS "REJECTED". DRAWINGS WHICH ARE OF NO RELEVANCE TO A STRUCTURAL REVIEW WILL BE RETURNED AS "NO STRUCTURAL RELEVANCE - NOT REVIEWED".

RESPONSIBILITY FOR THE ADEQUACY AND SAFE OPERATION OF SUCH EQUIPMENT, ITS APPENDAGES, ANCHORAGES AND VARIOUS SUPPORTS MECHANISMS WHETHER APPLIED TO OR EMBEDDED WITHIN THE STRUCTURAL BUILDING FRAMEWORK IS THE SOLE RESPONSIBILITY OF ITS DESIGNER OR RELATED CONTRACTOR, AND IS NOT THE RESPONSIBILITY OF QUINN DRESSSEL ASSOCIATES.

B/ GENERAL

- THE FOLLOWING GENERAL NOTES AND TYPICAL DETAILS SHOWN ON THE DRAWINGS ARE APPLICABLE TO ALL STRUCTURAL CONDITIONS NOT SPECIFICALLY DETAILED ON THE STRUCTURAL DRAWINGS.
- THE STRUCTURAL DRAWINGS SHALL BE READ WITH THE SPECIFICATIONS AND OTHER CONTRACT DOCUMENTS. REFER TO THE ARCHITECTURAL DRAWINGS FOR THE SEPARATION OF REQUIREMENTS.
- ONLY DIMENSIONS RELEVANT TO STRUCTURAL ELEMENTS ARE SHOWN ON THE STRUCTURAL DRAWINGS. FOR OTHER DIMENSIONS, AND LOCATIONS, SIZES OF ELEVATOR OR FORMER OPENINGS NOT SHOWN, REFER TO OTHER CONTRACT DRAWINGS AND DOCUMENTS. OBTAIN THE ARCHITECT'S PRIOR APPROVAL BEFORE INSTALLING SUCH OPENINGS, SLEEVES ETC.
- SEE OTHER CONTRACT DRAWINGS FOR LOCATIONS AND DIMENSIONS OF PTES, BASES, SLUMPS, TRENCHES, DEPRESSIONS, GROOVES, AND CHAMFERS NOT SHOWN ON STRUCTURAL DRAWINGS.
- THE DESIGN LIVE LOAD SHOWN ON THE DRAWINGS SHALL NOT BE EXCEEDED DURING CONSTRUCTION. THE CONTRACTOR SHALL MAKE ADEQUATE PROVISIONS FOR CONSTRUCTION LOADS AND FOR ADEQUATE TEMPORARY BRACING TO KEEP ALL STRUCTURAL AND NON-STRUCTURAL ELEMENTS PLUMB AND IN TRUE ALIGNMENT AT ALL PHASES OF CONSTRUCTION.

D/ STANDARDS & DESIGN CRITERIA

GENERAL

THIS BUILDING HAS BEEN DESIGNED IN ACCORDANCE WITH THE FOLLOWING CODES AND STANDARDS:

IBC-REG. 330.12

NBC 2015

CSA-K23 SERIES

IMPORTANCE CATEGORY

THIS BUILDING HAS BEEN DESIGNED FOR AN IMPORTANCE CATEGORY OF NORMAL.

DEAD LOADS

THE BUILDING HAS BEEN DESIGNED TO RESIST THE FOLLOWING DEAD LOADS. SPECIFIC DESIGN VALUES ARE REFERENCED ON INDIVIDUAL FRAMING PLANS.

- BUILDING SELF-WEIGHT
- WEIGHT OF ALL MATERIAL, OF CONSTRUCTION INCORPORATED INTO THE BUILDING AND PERMANENTLY SUPPORTED THEREON
- WEIGHT OF PARTITIONS, WHERE APPLICABLE
- VERTICAL GRAVITY LOADS DUE TO SOIL, PLANTING MATERIAL AND TREES

LIVE LOADS

THE BUILDING HAS BEEN DESIGNED TO RESIST LIVE LOADS DUE TO USE AND OCCUPANCY. SPECIFIC DESIGN VALUES ARE REFERENCED ON INDIVIDUAL FRAMING PLANS. LIVE LOAD REDUCTIONS AS PERMITTED UNDER THE APPLICABLE GOVERNING CODE HAVE BEEN INCORPORATED INTO THE ELEMENT DESIGN UNLESS APPROPRIATE.

SNOW & RAIN

THE BUILDING'S ROOFS HAVE BEEN DESIGNED TO RESIST LOADS DUE TO SNOW AND RAIN AS DEFINED BY THE APPLICABLE CODE. THE FOLLOWING FACTORS ARE UNIQUE TO THIS BUILDING:

$$\begin{aligned} &W(\text{IMPORTANCE FACTOR}) = 1.00 \text{ (U.S.)} \\ &S_0 \text{ (1 IN 50 YEAR GROUND SNOW LOAD)} = 0.80 \text{ (SLS)} \\ &S_0 \text{ (1 IN 50 YEAR ASSOCIATED RAIN LOAD)} = 0.7 \text{ kN/m}^2 \end{aligned}$$

THE SNOW LOADS APPLIED TO THE FOUNDATION ARE PROVIDED BY U-BUILD ON DRAWING 9006S-SHEET-1A DATED APRIL 05, 2024. ALL SNOW LOADS ARE PROVIDED BY U-BUILD.

WIND

THE BUILDING HAS BEEN DESIGNED TO RESIST WIND FORCES AS CALCULATED IN ACCORDANCE WITH THE FOLLOWING FORMULA:

$$P = 1wqGqf$$

$$\begin{aligned} \text{WHERE } w(\text{IMPORTANCE FACTOR}) &= 1.00 \text{ (U.S.) (TABLE 4.1.7.1)} \\ q \text{ (1 IN 50 YEAR REFERENCE VELOCITY PRESSURE)} &= 0.75 \text{ (SLS) (TABLE 4.1.7.1)} \\ G(\text{BASED UPON OPEN/ROUGH TERRAIN}) &= 0.41 \text{ kPa} \\ G_0(\text{GUST EFFECT FACTOR}) \cdot G_1(\text{EXTERNAL PRESSURE COEFFICIENT}) &= 0.10/2 \pm 1.0 \\ &2.0 \end{aligned}$$

THE WIND FORCES APPLIED TO THE FOUNDATION ARE PROVIDED BY U-BUILD ON DRAWING 9006S-SHEET-1A DATED APRIL 05, 2024. ALL WIND LOADS ARE PROVIDED BY U-BUILD.

EARTHQUAKE

SITE PROPERTIES:

SITE CLASS D - ASSUMPTION

$$\begin{aligned} S_{w2}(2) &= 0.149 \\ S_{w2}(3) &= 0.180 \\ S_{w2}(4) &= 0.080 \\ S_{w2}(5) &= 0.031 \\ S_{w2}(6) &= 0.0078 \\ S_{w2}(7) &= 0.0024 \\ F_0 &= 1.24 \text{ (Table 4.1.8.4.B)} \\ F_0 &= 1.55 \text{ (Table 4.1.8.4.D)} \\ PGA &= 0.091 \end{aligned}$$

IMPORTANCE CATEGORY:

NORMAL (I=1.0 (U.S.))

INTERSTOREY DRIFT RATIO LIMITED TO 0.0026

STRUCTURAL CONFIGURATION:

IN ACCORDANCE WITH TABLE 4.1.8.6, THE BUILDING'S STRUCTURAL CONFIGURATION IS JUDGED TO BE "REGULAR" IN THE FOLLOWING CATEGORY(ES) TYPE:

(R)REG(2)

= 0.185

BUILDING HEIGHT = 7.14 m

THE SEISMIC FORCES APPLIED TO THE FOUNDATION ARE PROVIDED BY U-BUILD ON DRAWING 9006S-SHEET-1A DATED APRIL 05, 2024. ALL SEISMIC LOADS ARE PROVIDED BY U-BUILD.

H/ CONCRETE AND REINFORCING

- CONCRETE COVER TO REINFORCING:
 - ALL CONCRETE POURED AGAINST EARTH: 75mm
 - ALL CONCRETE POURED IN FORMS, BUT EXPOSED TO EARTH OR WEATHER: BARS LARGER THAN 10M IN WALLS AND SLABS OR MAIN BARS IN BEAMS AND COLUMNS: 50mm BARS 10M AND SMALLER: 40mm
 - ALL CONCRETE POURED ON MUD SLAB: 40mm
 - CONCRETE NOT EXPOSED TO EARTH OR WEATHER: SLABS AND WALLS: 25mm SHEAR WALLS: 50mm BEAMS AND COLUMNS TO STEELUPS OR TIES: 40mm ALL CONCRETE SLABS EXPOSED TO DE-ICING SALTS: 40mm
- TOP BARS: 40mm BOTTOM BARS: 30mm
- MINIMUM CLEAR DISTANCE BETWEEN PARALLEL BARS SHALL BE THE GREATEST OF THE FOLLOWING:
 - 1.5 BAR DIAMETER
 - 1.10 TIMES MAXIMUM SIZE OF AGGREGATE
 - 25mm MINIMUM
- MINIMUM LAPS OF CONTINUOUS DEFORMED BARS SHALL BE 34 Ø OR 450mm WHICHEVER IS GREATER. LAPS OF WELDED WIRE FABRIC SHALL BE 102mm AT ENDS AND EDGES. INCREASE LAPS BY 20% FOR EPOXY COATED BARS. ALL DOWELS SHALL HAVE A MINIMUM EMBEDMENT OF 34 Ø OR 450mm MINIMUM, EXCEPT THAT DOWELS TO SLAB SHALL HAVE A MINIMUM EMBEDMENT OF 90mm. INCREASE LAP LENGTHS BY 20% FOR EPOXY COATED BARS.

KEY PLAN

REVISIONS / ISSUED TO

NO.	DATE	BY
1	ISSUED FOR COORDINATION	2024-05-14
2	ISSUED FOR PERMIT	2024-05-30
3	ISSUED FOR UPDATES	2024-07-09

ARCHITECT / CONSULTANTS

CONTRACTOR SHALL CHECK ALL DIMENSIONS ON THE WORK AND REPORT ANY DISCREPANCY TO THE CONSULTANT BEFORE PROCEEDING. THIS DRAWING IS NOT TO BE USED FOR CONSTRUCTION UNLESS AUTHORIZED IN WRITING BY CONSULTANT.

SEAL / SIGNATURE

ISSUED FOR CONSTRUCTION DATE

CLIENT NAME

ALOCON CONCRETE
FLOOR FINISHING
LTD.

DRAWING TITLE
AGGREGATES
PROCESS BUILDING
FOUNDATION

GENERAL NOTES

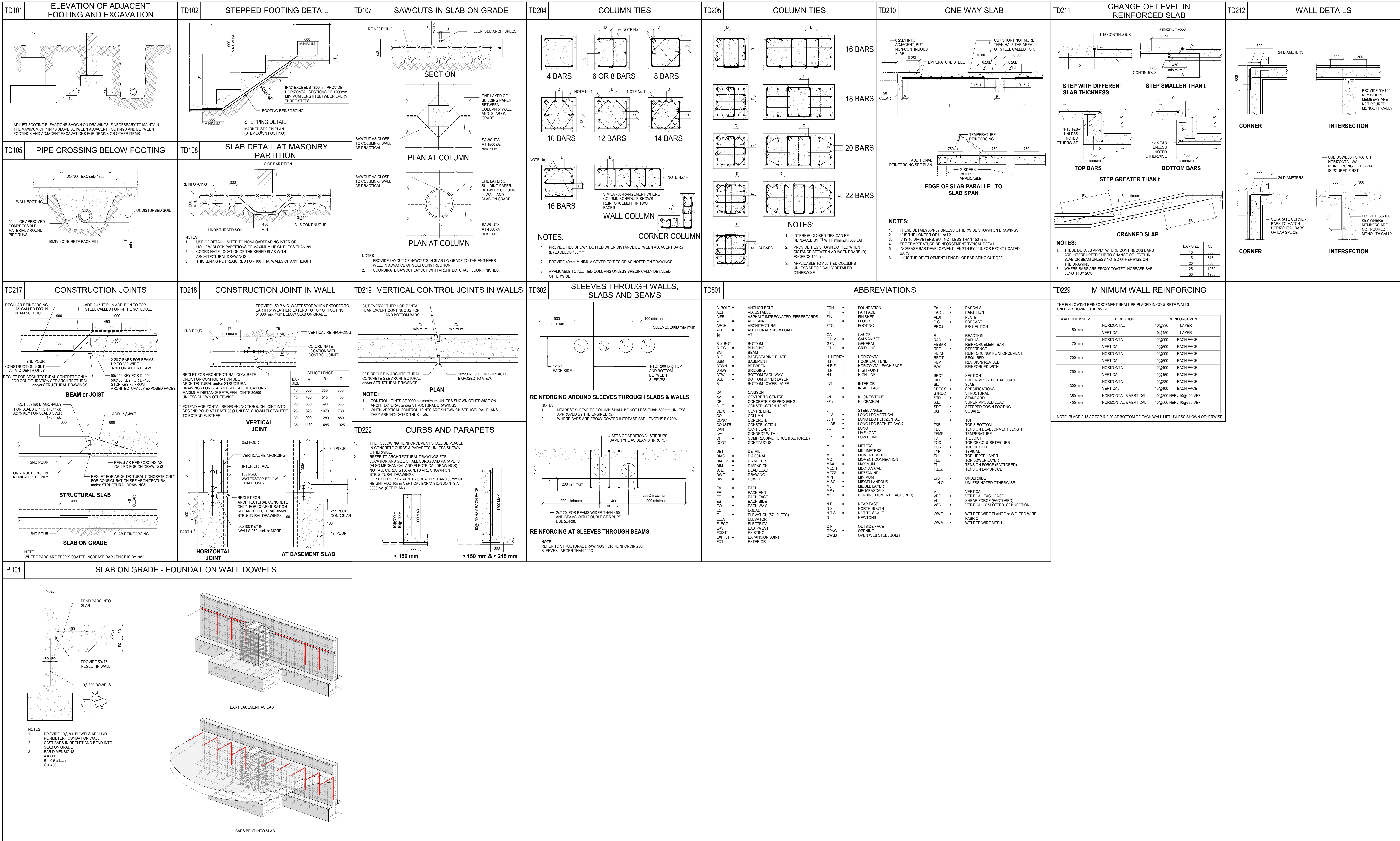
DESIGNED BY / DRAWN BY **Author**

SCALE **1 : 100**

DRAWING NUMBER **S101**

PROJECT NUMBER **24-060**

REVISIONS / ISSUED TO			
NO.	ISSUED FOR	DATE	BY
1	ISSUED FOR COORDINATION	2024-05-14	
2	ISSUED FOR PERMIT	2024-05-30	
3	ISSUED FOR UPDATES	2024-07-09	



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CLIENT NAME

ALOCON CONCRETE FLOOR FINISHING LTD.

DRAWING TITLE
AGGREGATES PROCESS BUILDING FOUNDATION

TYPICAL DETAILS

DESIGNED BY / DRAWN BY Author

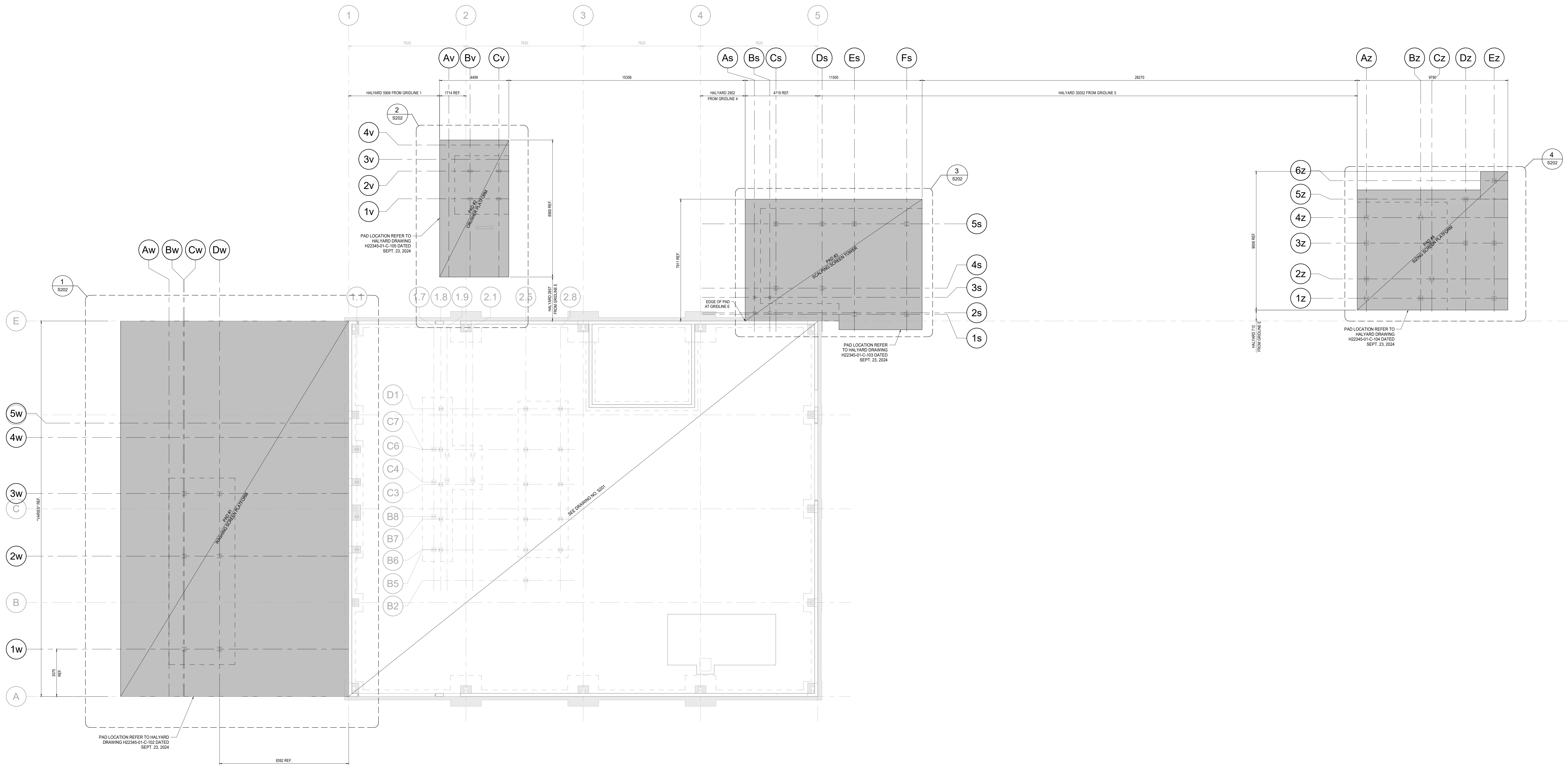
SCALE 1: 100

DRAWING NUMBER S102

PROJECT NUMBER 24-060

REVISIONS / ISSUED TO

NO.	DATE	BY
1	ISSUED FOR CLIENT REVIEW 2024-06-28	RY



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SEAL / SIGNATURE

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CLIENT NAME

ALOCON CONCRETE FLOOR FINISHING LTD.

DRAWING TITLE
AGGREGATES PROCESS BUILDING FOUNDATION

KEY PLAN

DESIGNED BY / DRAWN BY RY/FB

SCALE 1:100

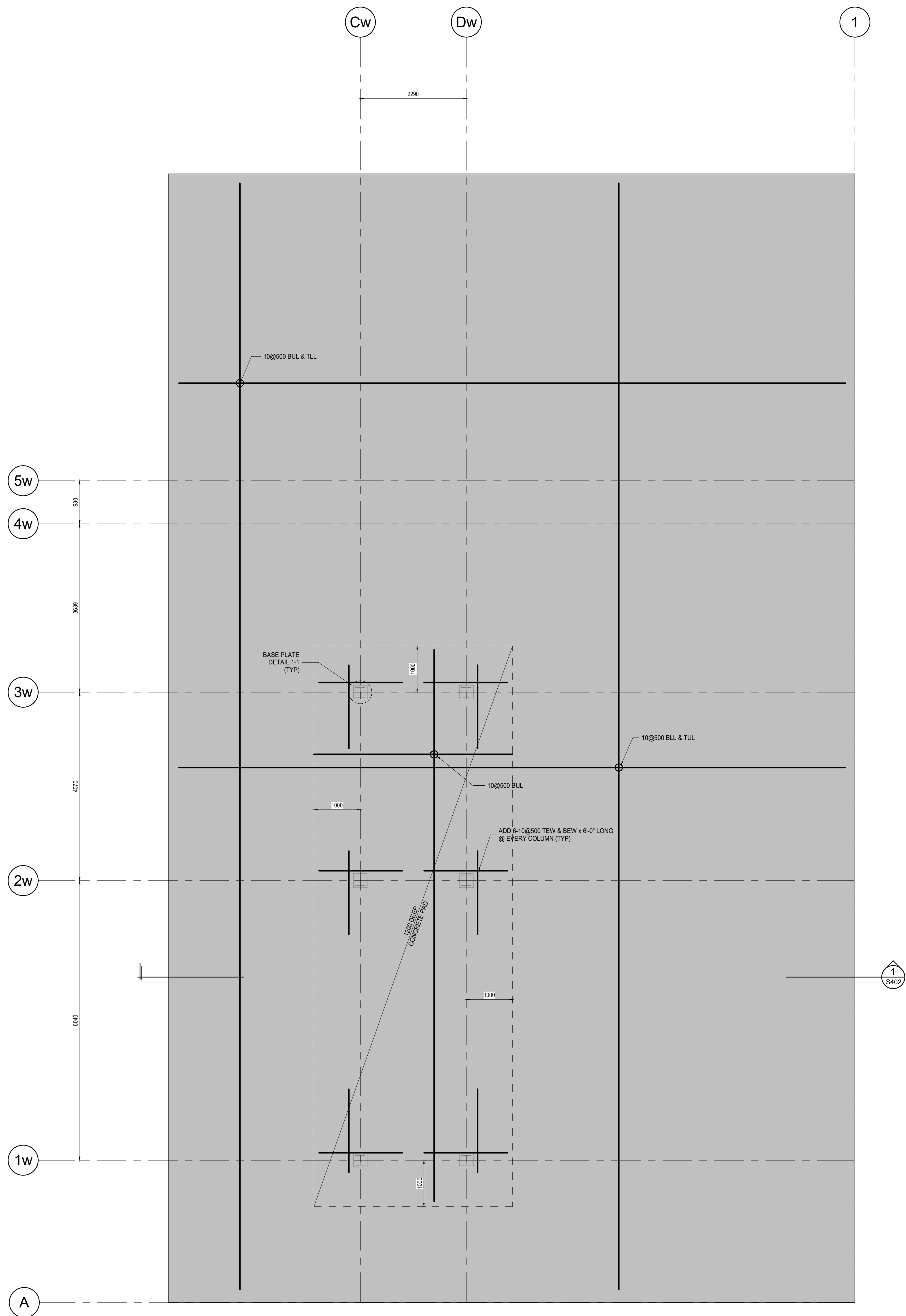
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S200

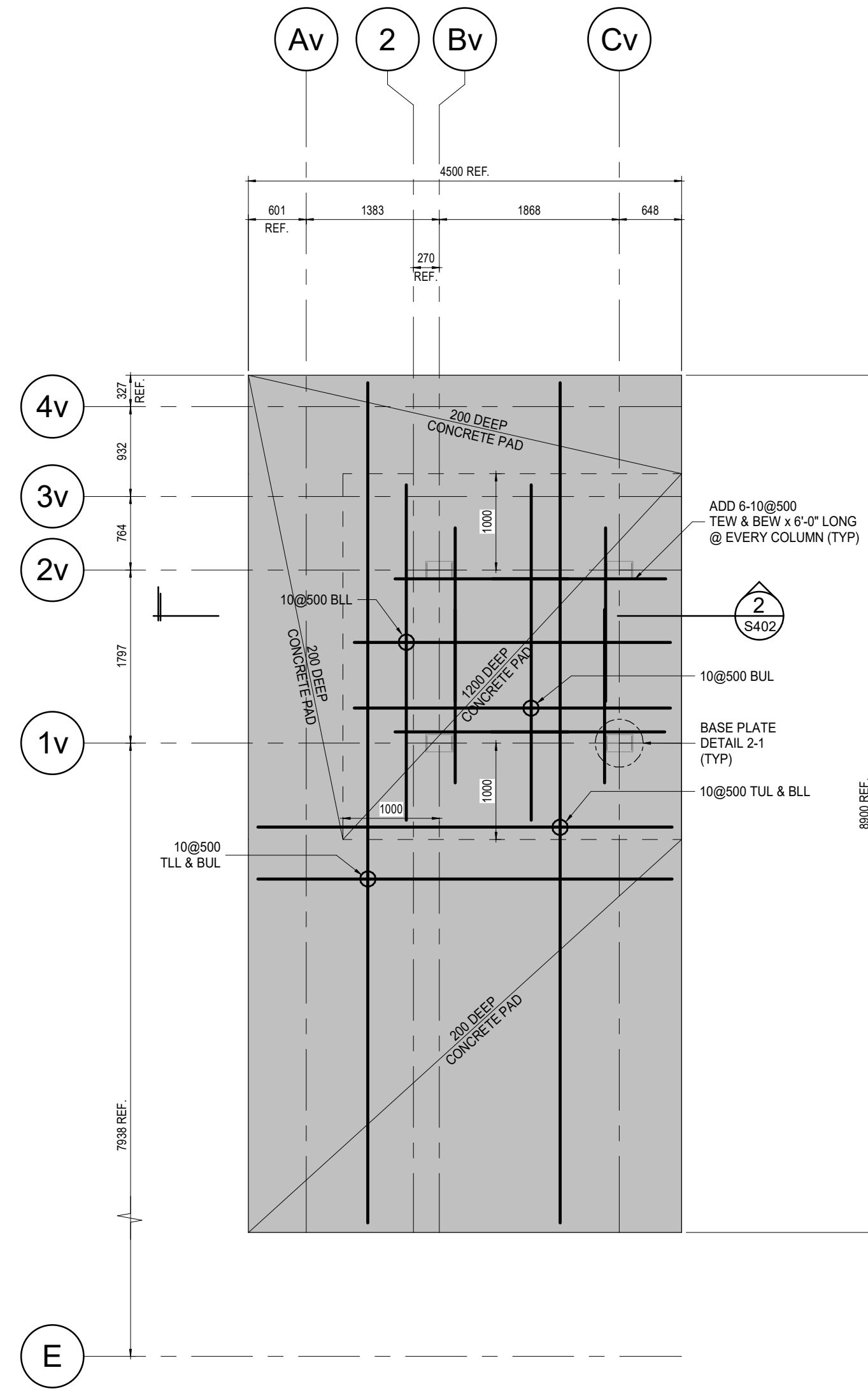
PROJECT NUMBER 24-060

REVISIONS / ISSUED TO

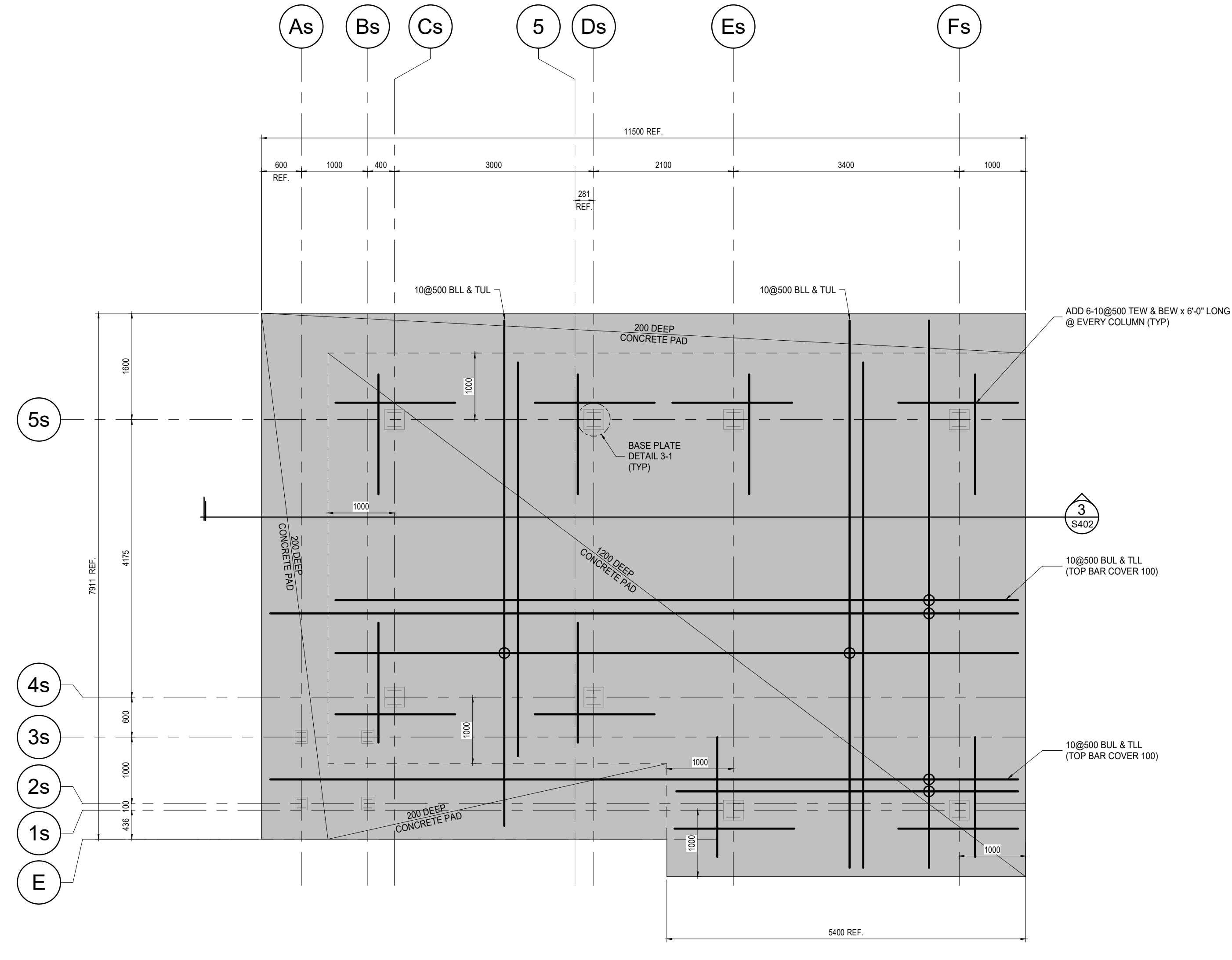
NO.	DATE	BY
1	ISSUED FOR CLIENT REVIEW 2024-09-17	RY



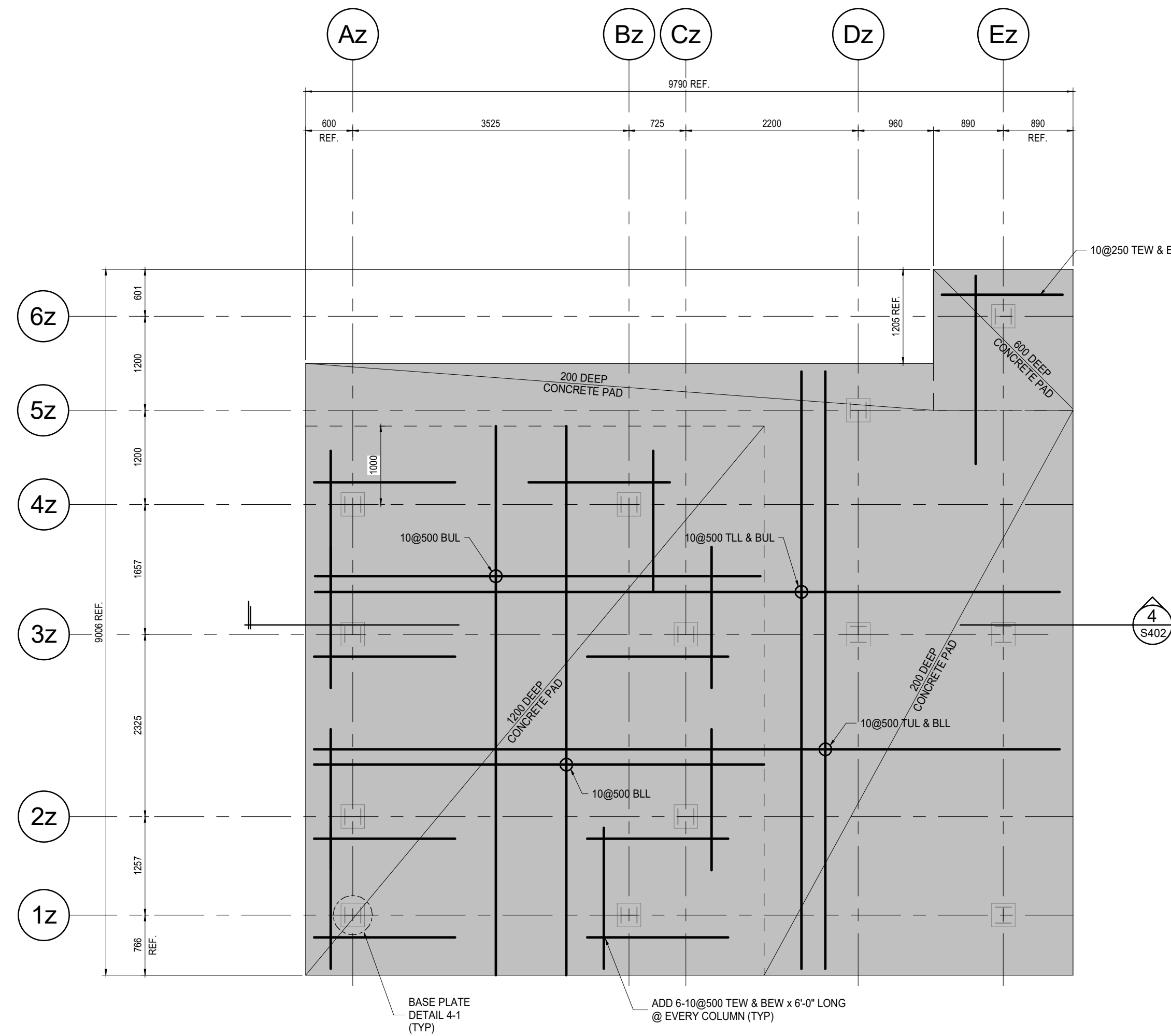
1 PAD #1 WASHING SCREEN PLATFORM
S202 1 : 50 $f_c = 35 \text{ MPa}$



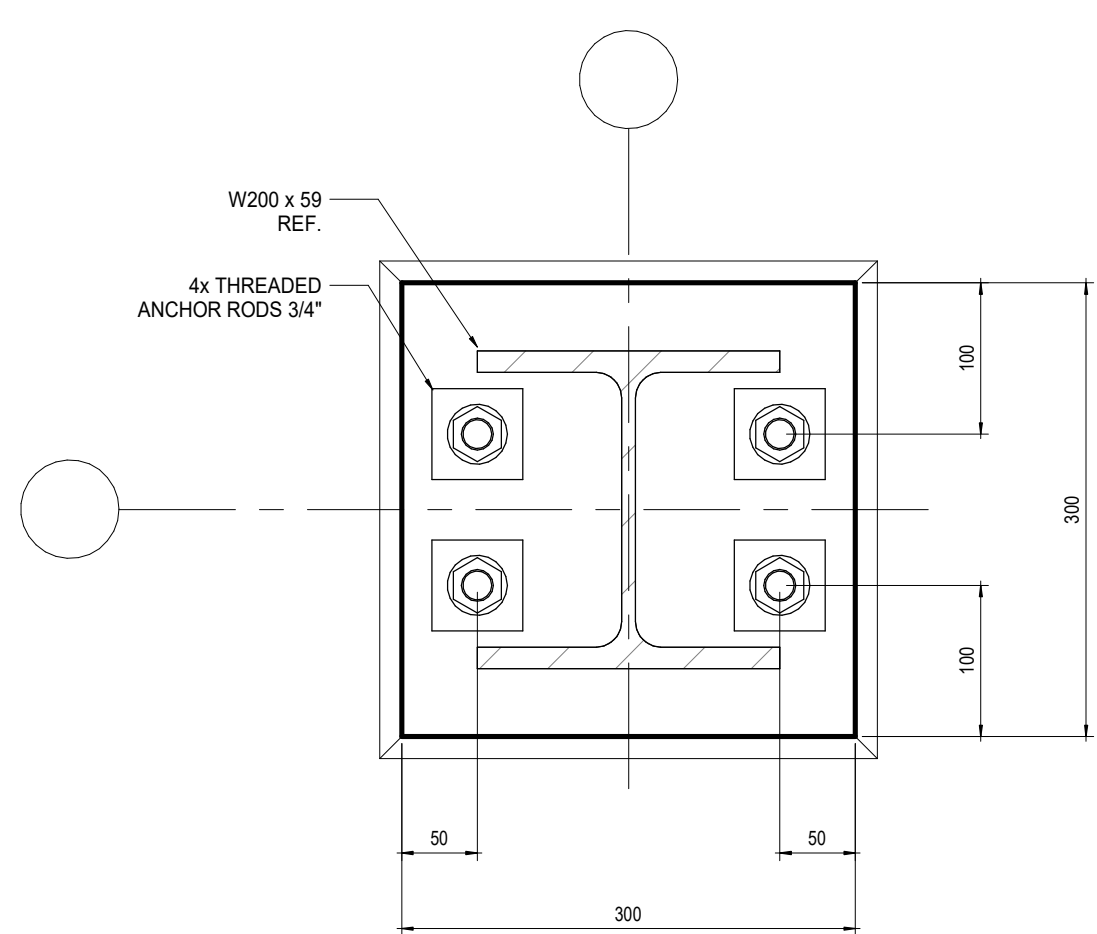
2 PAD #2 CRUSHER PLATFORM
S202 1 : 50 $f_c = 35 \text{ MPa}$



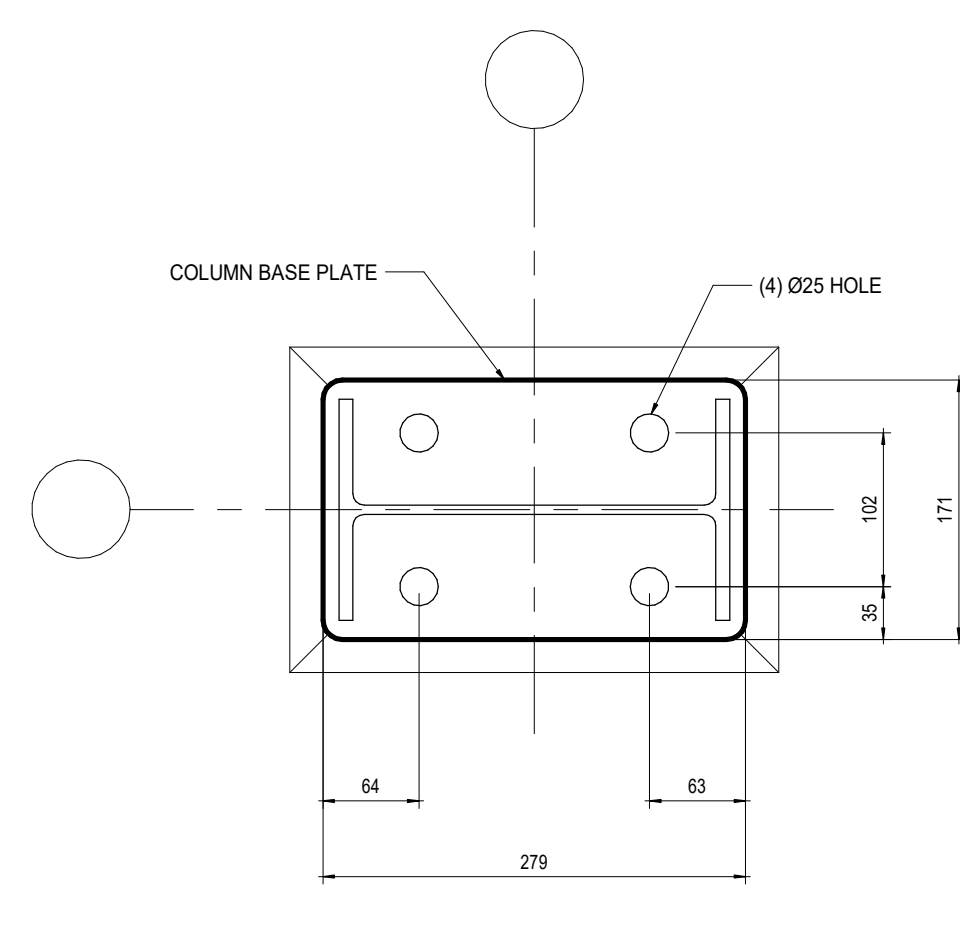
3 PAD #3 SCALPING SCREEN TOWER
S202 1 : 50 $f_c = 35 \text{ MPa}$



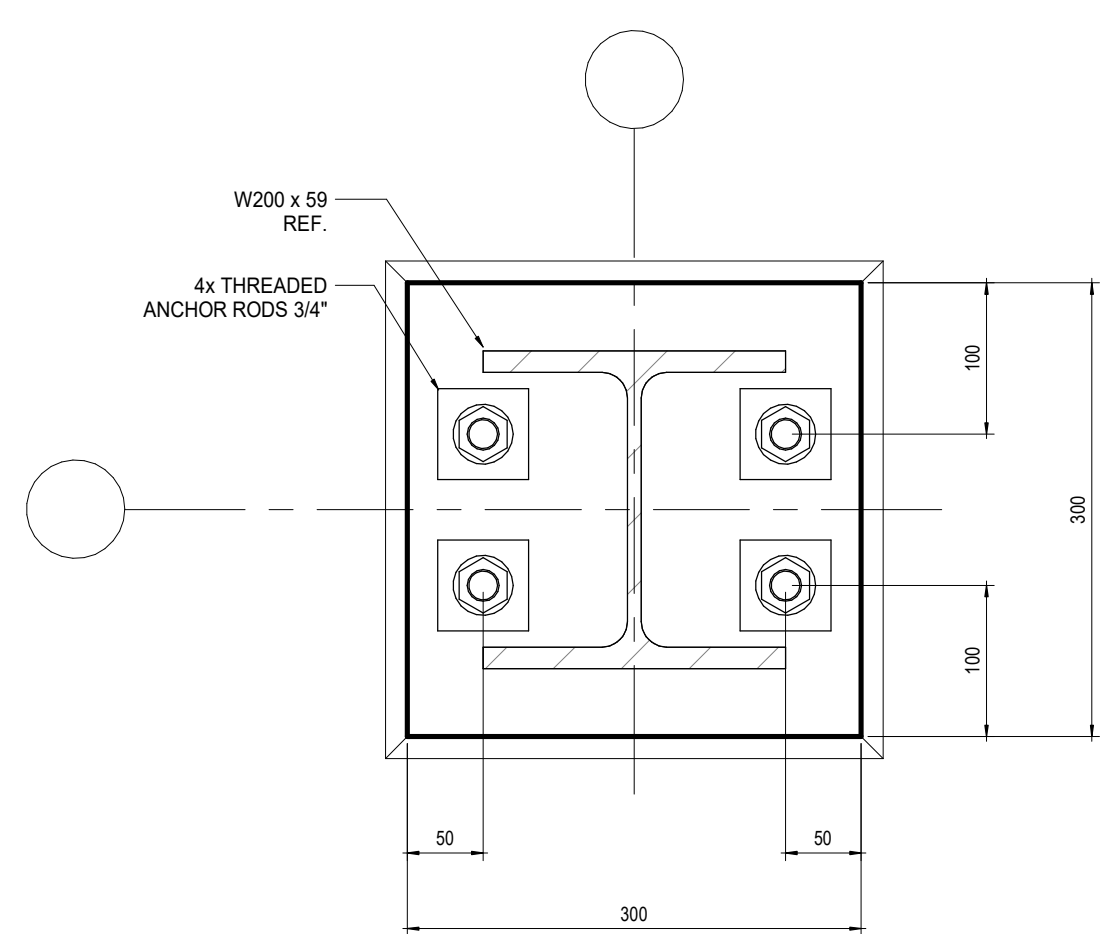
4 PAD #4 SIZING SCREEN TOWER
S202 1 : 50 $f_c = 35 \text{ MPa}$



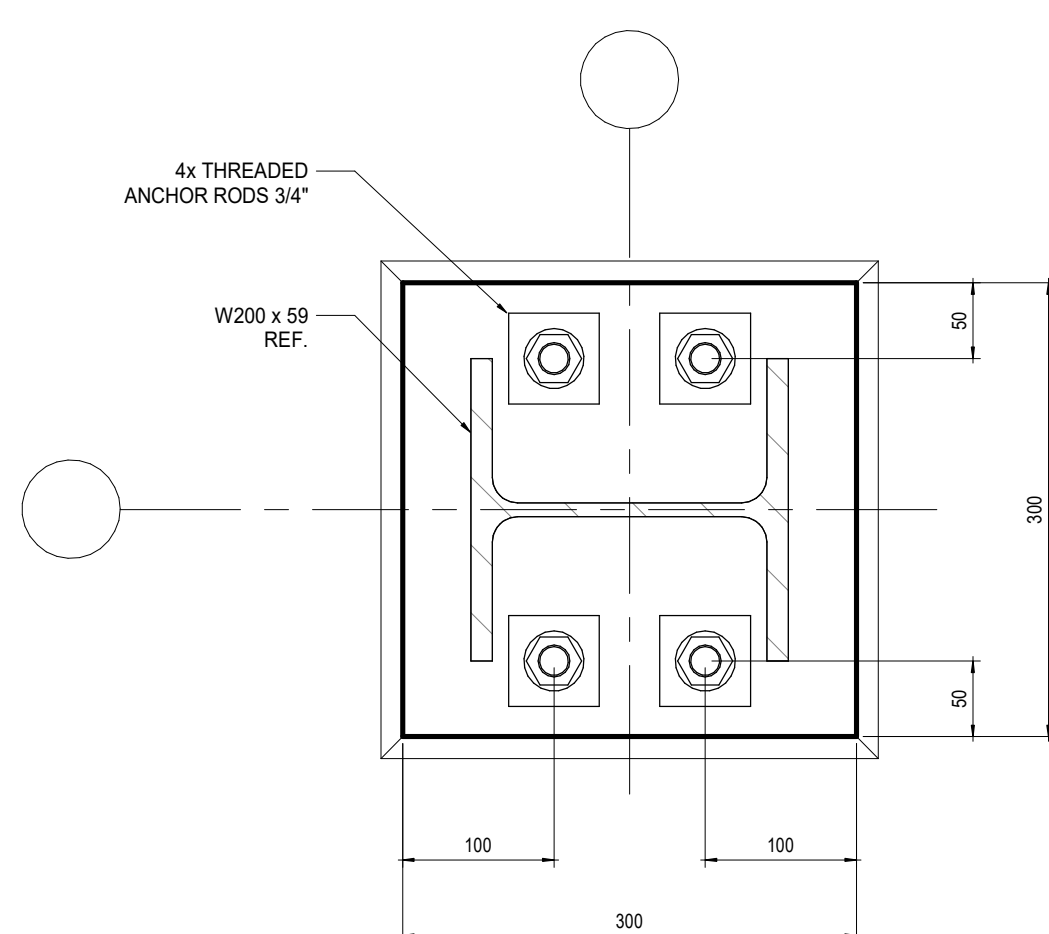
BASE PLATE DETAIL 1-1
1 : 5



BASE PLATE DETAIL 2-1
1 : 5



BASE PLATE DETAIL 3-1
1 : 5



BASE PLATE DETAIL 4-1
1 : 5

ARCHITECT / CONSULTANTS

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SEAL / SIGNATURE

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CLIENT NAME

ALOCON CONCRETE
FLOOR FINISHING
LTD.

DRAWING TITLE
AGGREGATES
PROCESS BUILDING
FOUNDATION

EXTERIOR STRUCTURE
FOUNDATION PLAN

DESIGNED BY / DRAWN BY RY/FB

SCALE As indicated

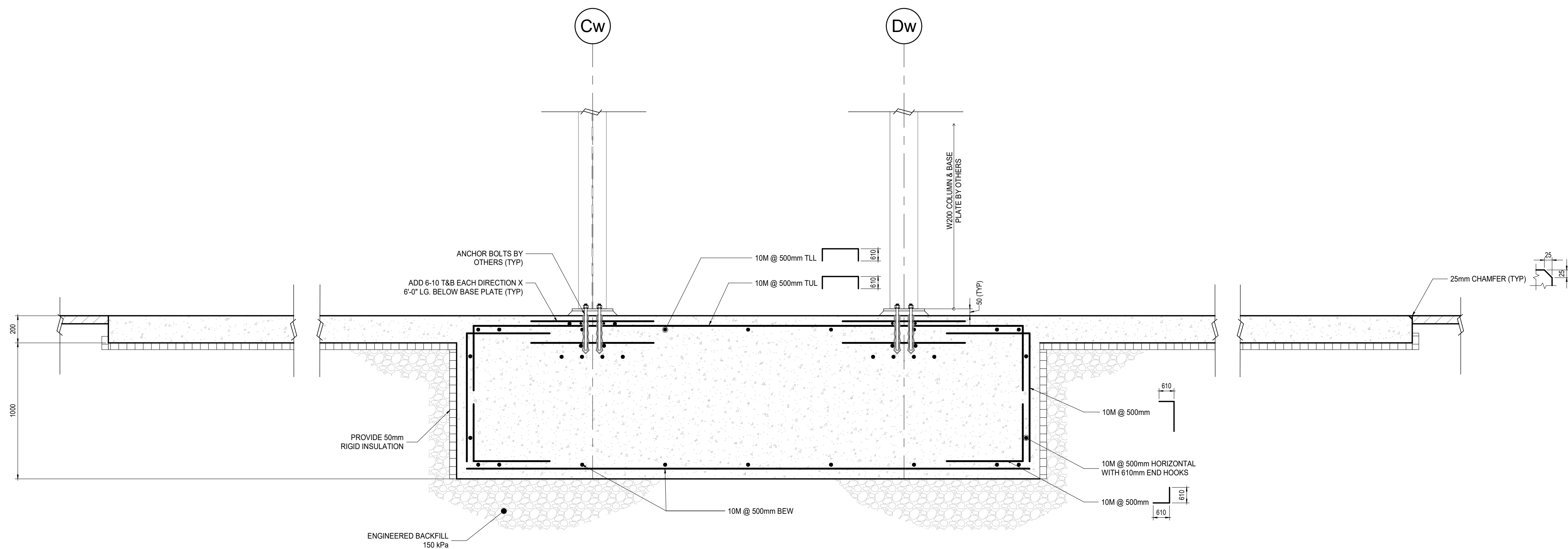
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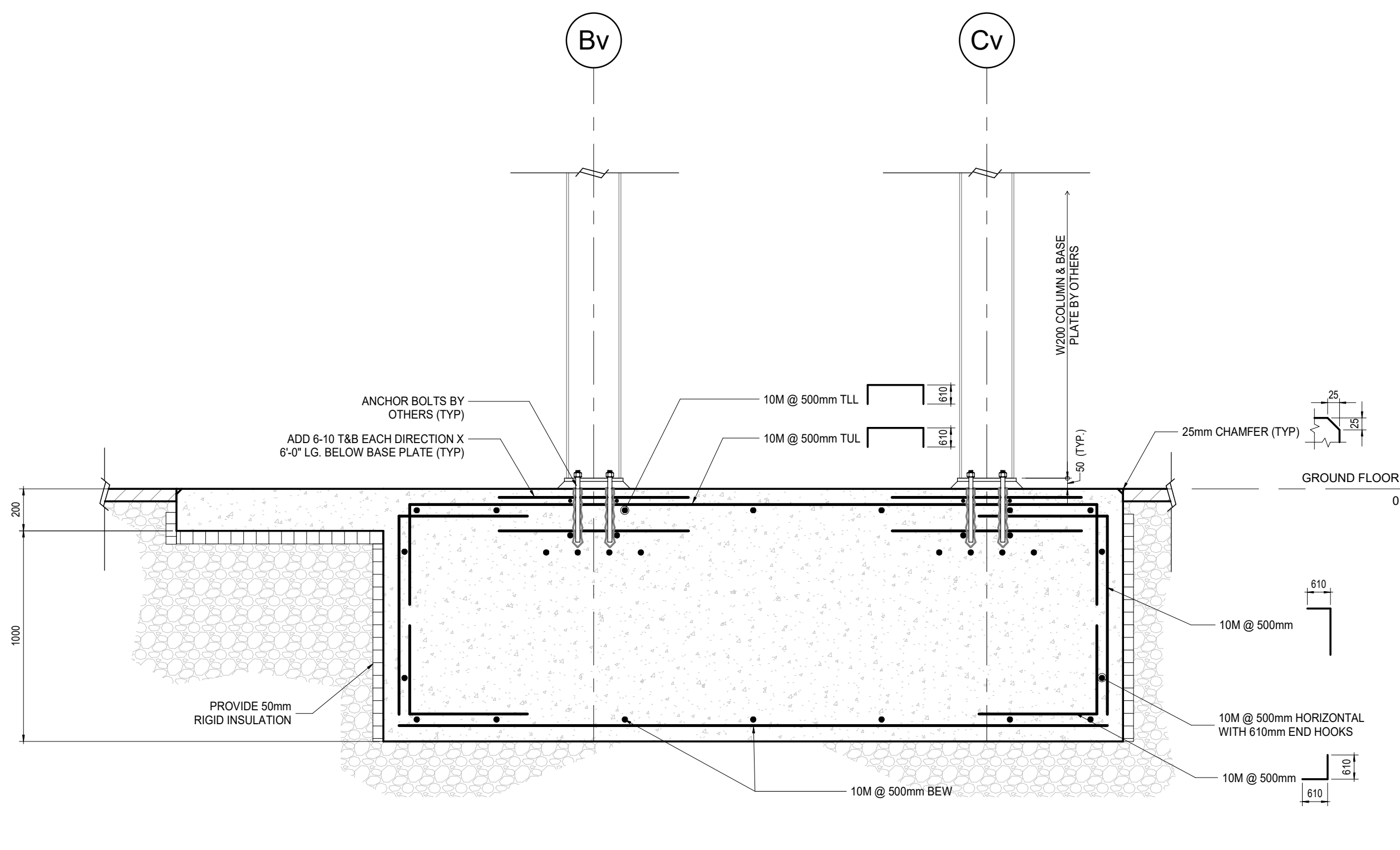
PROJECT NUMBER 24-060

REVISIONS / ISSUED TO

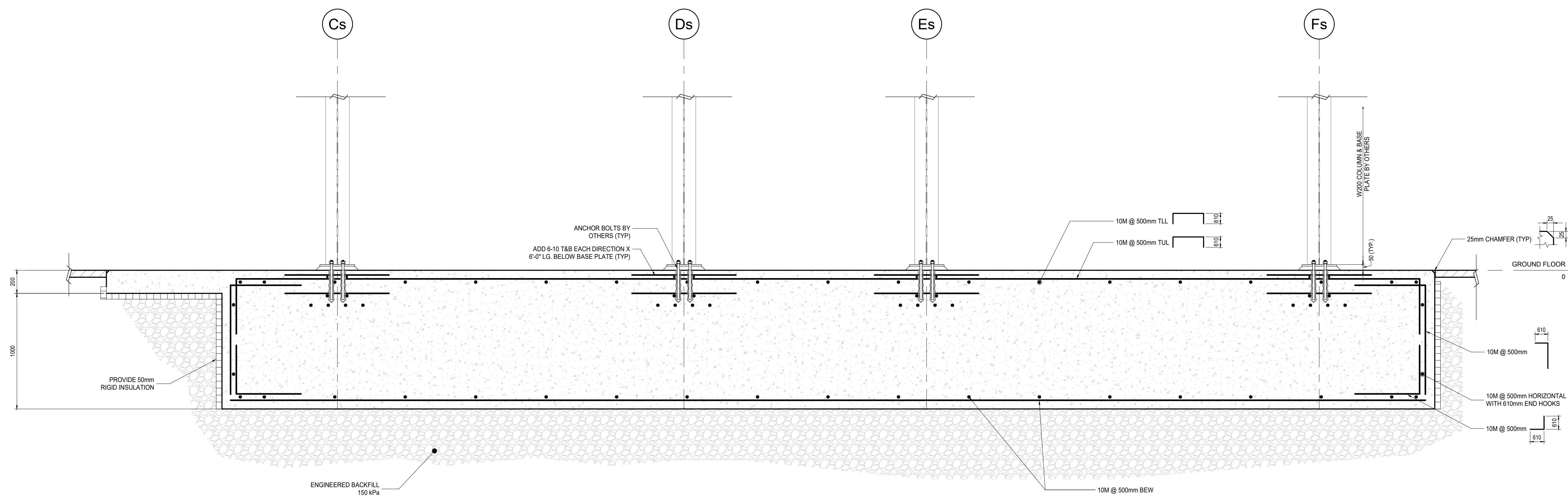
NO.	DATE	BY
1	ISSUED FOR CLIENT REVIEW 2024-09-17	RY



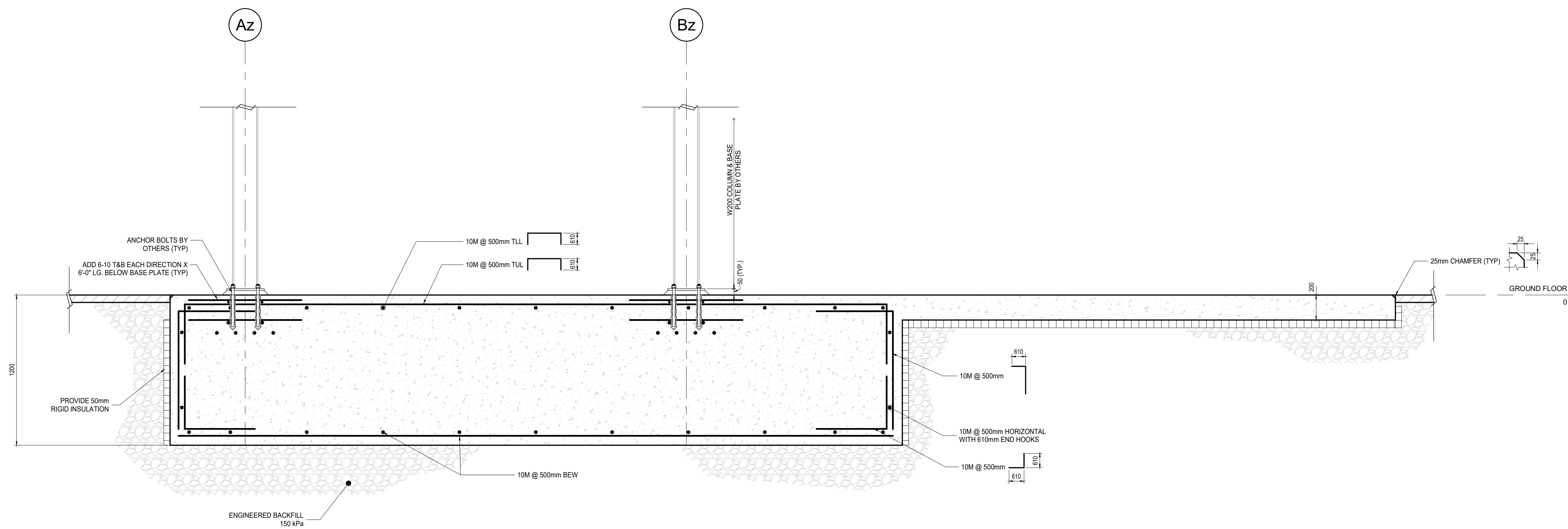
1 SECTION
S402 1 : 20



2 SECTION
S402 1 : 20



3 SECTION
S402 1 : 20



4 SECTION
S402 1 : 20

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FLOOR FINISHING
LTD.

DRAWING TITLE
AGGREGATES
PROCESS BUILDING
FOUNDATION

EXTERIOR STRUCTURE
FOUNDATION SECTIONS

DESIGNED BY / DRAWN BY

RY/FB

SCALE

1 : 20

DRAWING NUMBER

S402

PROJECT NUMBER

24-060